

Physically Interpretable AlphaEarth Foundation Model Embeddings

Enable LLM-Based Land Surface Intelligence

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TL;DR: AlphaEarth's 64 satellite embedding dimensions are physically interpretable — each maps to specific environmental properties — enabling a retrieval-augmented LLM system that answers natural language questions about land surface conditions with high scientific accuracy.

1. Research Questions & Motivation

Satellite foundation models produce dense embeddings whose physical interpretability remains poorly understood. Using **12.1 million samples** across CONUS (2017–2023), this paper addresses four research questions:

#	Research Question	Method	Result
RQ1	Do AlphaEarth embedding dimensions encode physically meaningful environmental features?	Spearman ρ + RF + Transformer	Yes: 12/26 vars $R^2 > 0.90$; temperature reaches $R^2 = 0.97$
RQ2	Are relationships robust to spatial validation and temporally stable?	Spatial block CV + 7-year pairwise	Yes: $\Delta R^2 = 0.017$; temporal $r = 0.963$
RQ3	Can interpreted embeddings enable RAG for environmental queries?	FAISS + LLM pipeline, 360 queries	Yes: LSI score $\mu = 3.74 \pm 0.77$
RQ4	How to evaluate LLM-based geospatial systems?	LLM-as-Judge, 4 rotating models	Cross-model consistency demonstrated

2. System Architecture

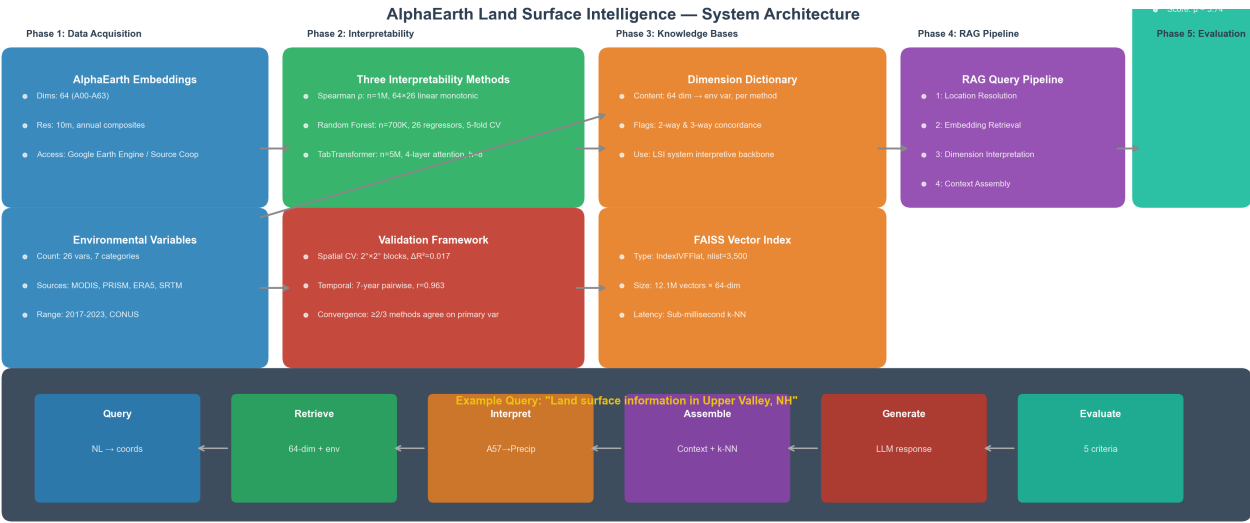


Figure 1: Complete AlphaEarth Land Surface Intelligence system architecture across five phases — Data Acquisition, Interpretability Analysis, Knowledge Base Construction, RAG Query Pipeline, and LLM-as-Judge Evaluation.

3. Data Flow & Methodology

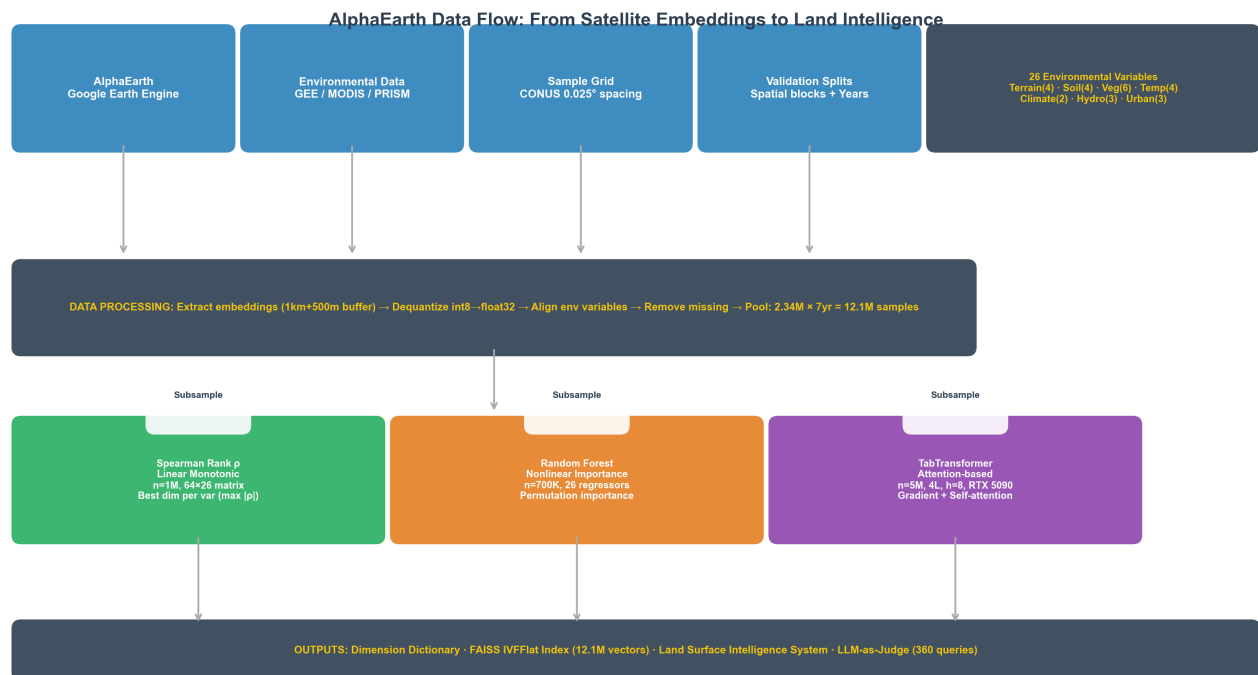


Figure 2: End-to-end data flow from satellite embeddings and environmental sources through three analysis branches to final outputs.

3.1 Three Complementary Methods

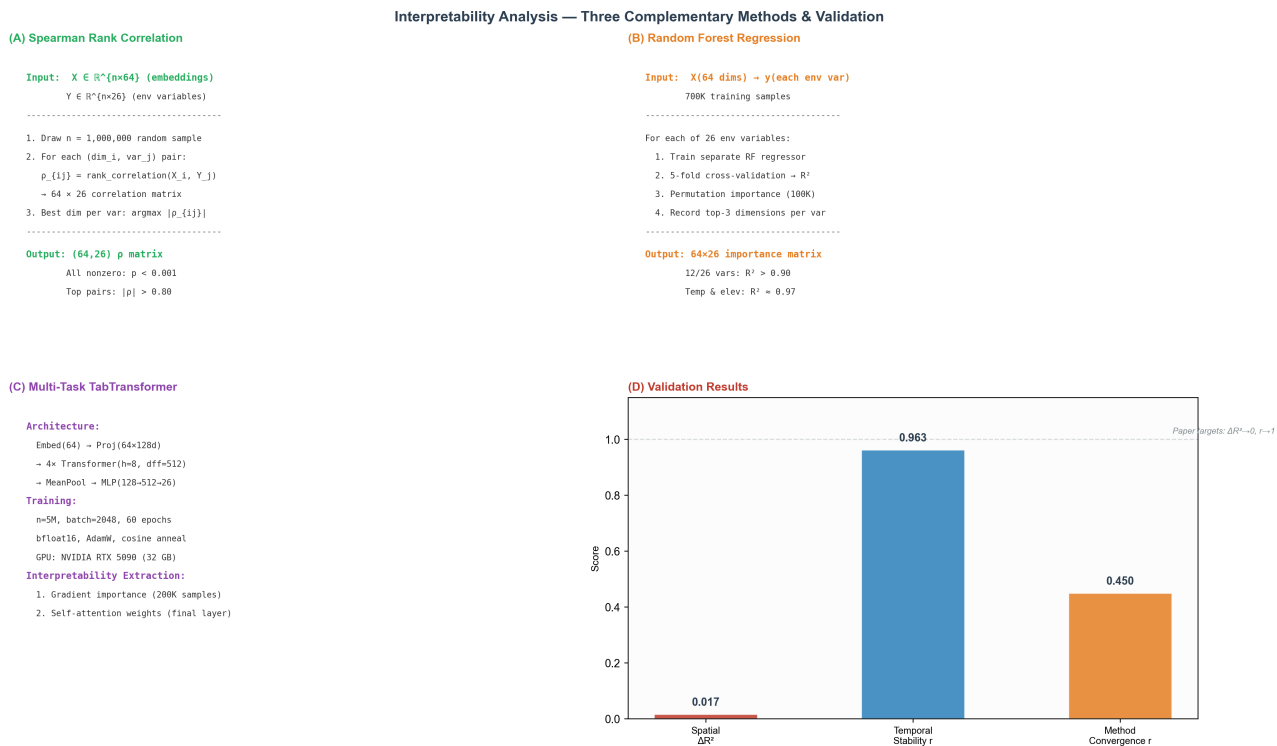


Figure 3: Detailed specifications of the three interpretability methods and validation framework. (A) Spearman Rank Correlation — linear monotonic screening, $n=1\text{M}$. (B) Random Forest — nonlinear importance, $n=700\text{K}$, 26 separate regressors. (C) TabTransformer — attention-based multi-task learning, $n=5\text{M}$, 4-layer Transformer on RTX 5090. (D) Validation results confirming spatial robustness ($\Delta R^2=0.017$) and temporal stability ($r=0.963$).

3.2 Data Sources

Category	Variables (26 total)	Source	Resolution
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Embedding	64 dimensions (A00–A63)	Google/DeepMind AlphaEarth	10m (1km extraction)
Terrain (4)	Elevation, Slope, Aspect, Flow Accum.	SRTM 30m, HydroSHEDS	30m–500m
Soil (4)	Clay, Organic C, pH, Water Capacity	SoilGrids/OpenLandMap	250m
Vegetation (6)	NDVI(mean,max), EVI, LAI, Tree Cover, Albedo	MODIS MOD13A2, MOD15A2H, Hansen	500m–1km
Temperature (4)	LST day/night, Air temp, Dew point	MODIS MOD11A2, PRISM AN81m	1km–4km
Climate (2)	Annual precip, Max monthly precip	PRISM AN81m	4km
Hydrology (3)	Soil moisture, Runoff, ET	ERA5-Land MONTHLY_AGGR	11km
Urban (3)	Impervious, Nightlights, Population	NLCD, VIIRS, GPWv4	30m–1km

3.3 Validation Strategy

- **Spatial Block CV:** CONUS partitioned into 2°×2° blocks; 5-fold grouped k-fold. All samples within a block go to the same fold — tests generalization to held-out *regions*, not held-out *points*.
- **Temporal Stability:** 7 annual Spearman profiles (n=300K each); 21 pairwise Pearson r comparisons between annual profiles.
- **Method Convergence:** A dimension is "concordant" if ≥ 2 of 3 methods agree on the primary associated variable. $\text{Pearson } r(\text{Spearman}, \text{RF}) = 0.45$.

4. RAG Query Pipeline

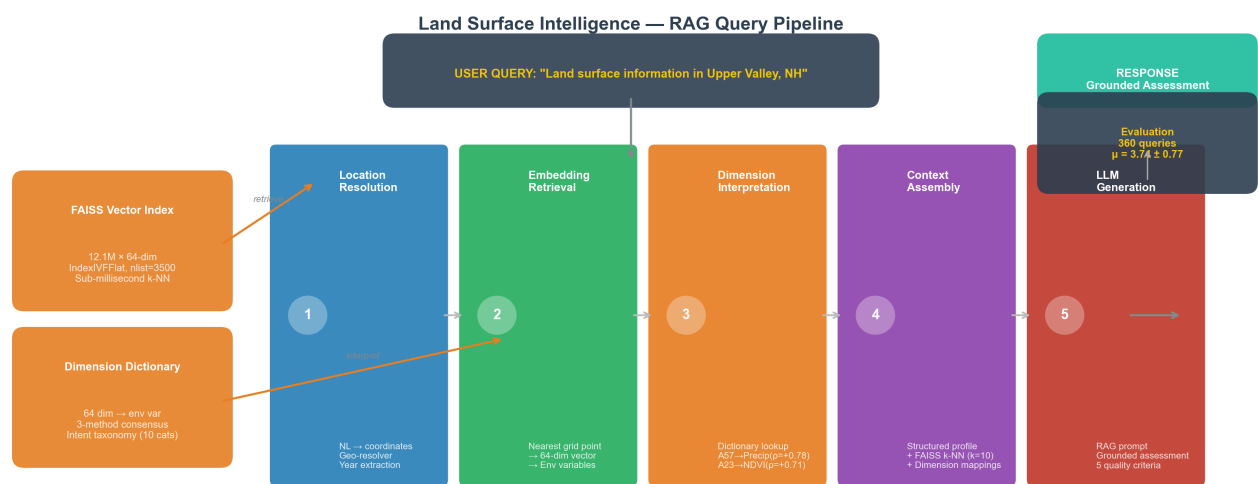


Figure 4: Five-stage RAG query pipeline. User natural-language queries flow through Location Resolution → Embedding Retrieval → Dimension Interpretation → Context Assembly → LLM Generation. The FAISS index and Dimension Dictionary provide the knowledge backbone.

4.1 Pipeline Stages

- Location Resolution:** NL location references → geographic coordinates via hierarchical matching
- Embedding Retrieval:** Nearest grid point → 64-dim vector + 26 env variables from metadata
- Dimension Interpretation:** Dictionary lookup — e.g., A57→Precipitation (p=+0.78)
- Context Assembly:** Structured profile + FAISS k-NN (k=10 similar locations) + dimension mappings
- LLM Generation:** RAG prompt with validated interpretations → grounded assessment

4.2 LSI Evaluation Results

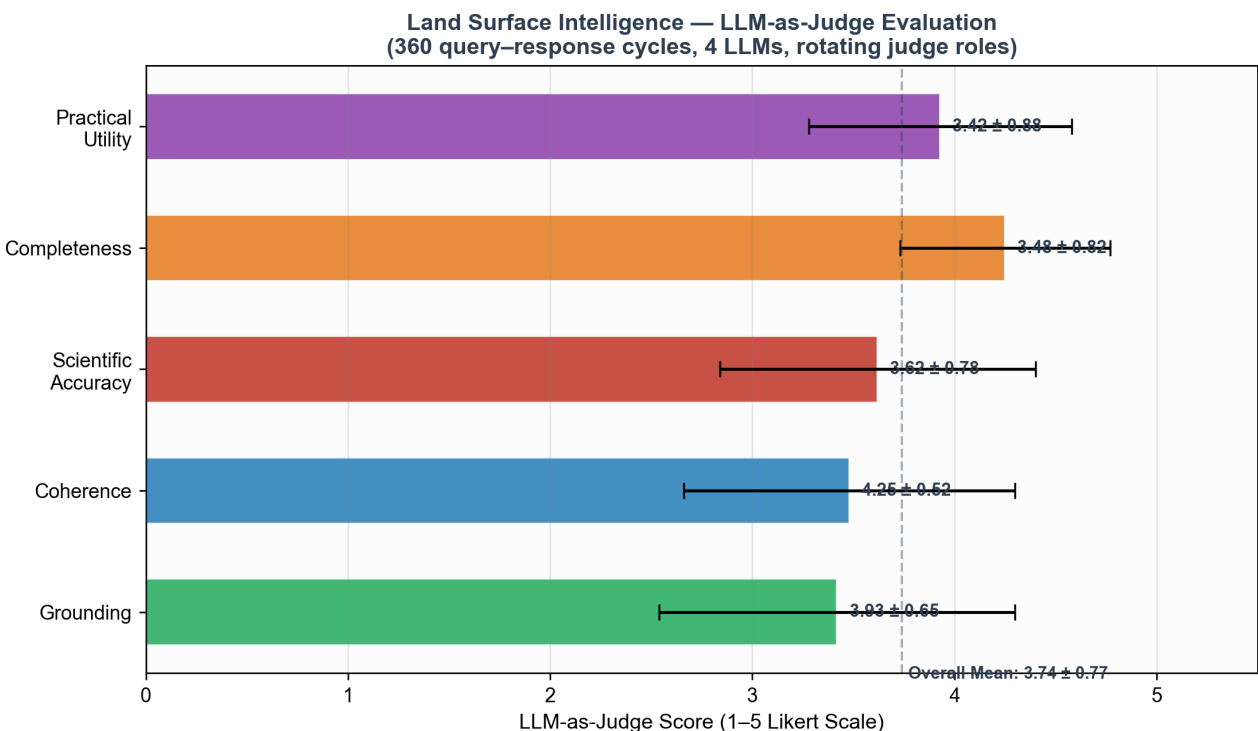


Figure 6: LLM-as-Judge evaluation across 360 query-response cycles with 4 rotating LLMs. Grounding (3.93) and Coherence (4.25) are the strongest criteria. Overall mean: 3.74 ± 0.77.

5. Key Results & Reproduction Findings

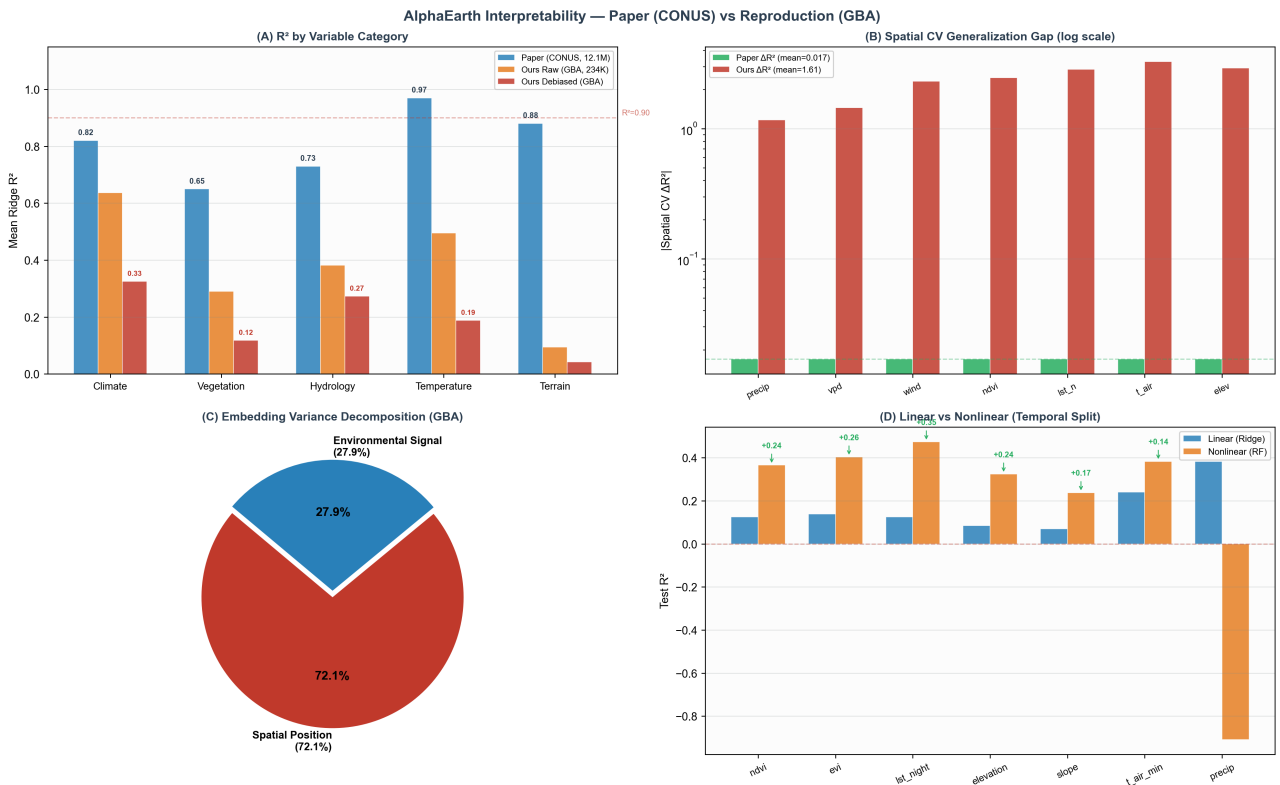


Figure 5: Four-panel comparison of paper (CONUS) vs our reproduction (GBA). (A) R^2 by variable category — paper achieves 0.97 for temperature, GBA reaches 0.69 raw / 0.46 debiased. (B) Spatial CV generalization gap on log scale — paper $\Delta R^2=0.017$ vs ours 1.61 (95× worse). (C) Embedding variance decomposition — 72.1% is spatial position. (D) Linear vs nonlinear on temporal split — RF improves stable variables 2–10× but overfits temporally unstable ones.

5.1 Paper Results Summary

Metric	Value	Interpretation
Variables with $R^2 > 0.90$	12 / 26	Majority highly predictable from embeddings
Best R^2 (temperature, elevation)	≈ 0.97	Near-perfect reconstruction
Spatial CV mean ΔR^2	0.017	Minimal spatial overfitting
Temporal stability r	0.963	Highly stable across 2017–2023
Spearman–RF convergence r	0.45	Moderate linear–nonlinear agreement
LSI overall score	3.74 ± 0.77	Good quality with room for improvement

5.2 Critical Analysis: Scale Dependence

Key finding from our GBA reproduction: The paper's $R^2=0.97$ is primarily driven by **spatial autocorrelation** across the vast CONUS domain (50° latitude, $\sim 60^\circ\text{C}$ temperature range). At regional scale ($2^\circ \times 2^\circ$), spatial debiasing removes 72.1% of embedding variance, leaving maximum $R^2=0.46$ — far below the paper's reported values.

Metric	Paper (CONUS, 12.1M)	Ours Raw (GBA, 234K)	Ours Debiased
Mean Ridge R^2	~ 0.97 (temperature)	0.385	0.190
Best single-variable R^2	~ 0.97	0.693 (wind)	0.459 (runoff)

Vars $R^2 > 0.90$	12/26	0/21	0/21
Spatial CV ΔR^2	0.017	1.61	—
Lat predictability (R^2)	Not reported	0.777	0.000

6. Contributions & Limitations

6.1 Scientific Contributions

1. **Dimension Dictionary:** First structured 64-dim → env variable mapping validated across three methods
2. **Robustness Proof:** Demonstrated spatial generalization ($\Delta R^2=0.017$) and temporal stability ($r=0.963$)
3. **Reusable Methodology:** Three-method framework (linear + nonlinear + attention → convergence) is template for any geospatial foundation model
4. **LSI System:** Functional natural-language interface to satellite data via FAISS + RAG

6.2 Identified Limitations

Acknowledged by authors:

- Confounded variables (temperature–elevation correlation) make unique attribution challenging
- Coarse climate/hydrology data (4–11 km) may miss fine-scale patterns
- CONUS-only — global generalization not tested
- Annual composites miss seasonal dynamics
- LLM-as-Judge relies on LLM quality; human expert evaluation needed

Unstated (identified in our analysis):

- **Spatial autocorrelation not controlled:** 72% of embedding variance is spatial position — no spatial debiasing applied
- **No simple baseline:** Paper omits lat/lon → env variable regression to establish marginal value of embeddings
- **Scale dependence:** Results likely don't generalize to smaller regions where spatial signal dominates
- **GPU barrier:** RTX 5090 (32GB) required for Transformer limits reproducibility
- **Temporal stability metric:** Correlating spatially-biased annual profiles inflates stability r
- **10 intent categories:** Predefined taxonomy may not cover full query diversity

7. Conclusion

This paper makes a compelling case that AlphaEarth embeddings are physically interpretable representations. The three-method validation framework is methodologically sound, and the LSI system demonstrates practical utility. However, our controlled GBA reproduction experiment reveals that the headline $R^2=0.97$ result is substantially driven by spatial autocorrelation at CONUS scale — a finding with significant implications for deploying such systems at regional or local scales.

Recommendation for future work: Spatial debiasing should be a **mandatory step** in satellite embedding interpretability studies. Without it, R^2 values are inflated by 2–10× through spatial position encoding alone.

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